

| Key Performance Parameter                      | Conventional EPS<br>(20% Replacement) | Upgraded EPS<br>(20% Replacement) | Conventional EPS<br>(40% Replacement) | Upgraded EPS<br>(40% Replacement) |
|------------------------------------------------|---------------------------------------|-----------------------------------|---------------------------------------|-----------------------------------|
| Density, 28-day Oven-Dry (kg/m³)               | 2011.0 ± 18.5                         | 1969.0 ± 17.0                     | 1748.0 ± 15.0                         | 1734.0 ± 14.5                     |
| Compressive Strength, f <sub>c</sub> (MPa)     | 10.15 ± 0.65                          | 15.62 ± 0.85 (+53.9%)             | 4.20 ± 0.35                           | 5.21 - 7.15 (+70.2%)              |
| Flexural Strength, f <sub>r</sub> (MPa)        | 2.10 ± 0.15                           | 3.59 ± 0.20 (+71.0%)              | 0.95 ± 0.08                           | 1.80 - 1.85 (+94.7%)              |
| Fracture Toughness (mJ/mm³)                    | 12.58 ± 0.85                          | 18.24 ± 1.20 (+45.0%)             | 1.87 ± 0.15                           | 3.84 ± 0.25 (+105.3%)             |
| Apparent Porosity (%)                          | 7.65 ± 0.25%                          | 6.48 ± 0.24%                      | 8.03 ± 0.30%                          | 6.94 ± 0.26%                      |
| Water Absorption (%)                           | 3.82 ± 0.15%                          | 3.34 ± 0.14%                      | 4.50 ± 0.18%                          | 3.52 ± 0.15%                      |
| ITZ Ca/Si Ratio (SEM-EDS)                      | 23.50 ± 2.40 (CH)                     | 6.80 ± 0.55 (C-S-H)               | 24.85 ± 2.70 (CH)                     | 5.65 ± 0.39 (-77.3%)              |
| Strength-to-Cost (MPa / kLKR·m <sup>-3</sup> ) | 0.275 ± 0.020                         | 0.323 ± 0.025 (+17.5%)            | 0.098 ± 0.009                         | 0.131 ± 0.011 (+33.7%)            |
| Embodied Carbon (kg CO <sub>2</sub> /MPa·m³)   | 53.47 ± 2.80                          | 32.29 ± 1.60 (-39.6%)             | 130.02 ± 6.50                         | 95.66 ± 4.50 (-26.4%)             |